

GHOJA: Human-in-the-Loop Joint Pose Optimization based on Geometric Constraint and Human Common-sense

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ABSTRACT

Assigning a motion to movable parts in 3D scanned objects is important for various applications, such as virtual reality and robotics simulation. We call this problem *joint-assignment* that requires determining the joint type, position, and orientation. Although a machine can calculate the joint pose that meets the geometric constraint by utilizing a 3D model, the calculation result is not uniquely determined. These results can be narrowed down uniquely based on human common-sense, such as recognizing object affordance. In this study, we propose a human-in-the-loop joint-assignment system, GHOJA (Geometric- and Human-based Optimization for Joint-Assignment). The system considers both geometric constraints and common-sense. The results of a user experiment demonstrate that GHOJA is significantly capable of satisfying the geometric constraint compared to the manual assignment by a human. In addition, this system significantly meets common-sense compared to the automatic optimization based on the geometric constraint.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

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1 INTRODUCTION

An emergence of accurate scanning devices [2] and deep learning-based 3D reconstruction techniques [4, 20] has brought attention to an application of reconstructed objects, such as an interaction in virtual reality (VR) [19] and robotics simulation [7, 22]. Since such 3D models represent a static state, an object with movable parts,

such as a slider and knob, lacks information about the movement of these parts. In other words, an *object joint* is lacking. This leads to a limited scope of applications for the 3D reconstruction. Assigning the joint to the 3D models, *i.e.*, *joint-assignment*, facilitates applications in a wide range of fields, including the interaction with objects in VR [19] and robot motion planning [7]. We aim to perform joint-assignment to the 3D models lacking the joint.

In joint-assignment, it is necessary to determine a type of joint and a position and orientation of the joint. There have been primarily two approaches to determining the joint type and pose. One approach involves manual assignment by a human using a user interface (UI) [16, 18, 19, 23], and the other involves estimation by a machine based on knowledge learned from extensive data [8, 11, 12, 17]. In the former, a human cannot determine the precise joint pose manually due to continuous pose values. The latter approach also faces the problem of insufficient estimation accuracy of the pose. For instance, in [11], an error of joint-orientation is approximately 20° in the real world. Insufficient accuracy leads to breaking a geometric constraint, such as an interpenetration and separation between object parts. Using the 3D objects in such a state for interaction in VR compromises a user experience. Furthermore, there is a risk of object destruction in robot motion planning. Therefore, satisfying the geometric constraint is critically important in such applications.

In fact, machines can automatically meet the constraint by leveraging the 3D model. For instance, by quantifying the geometric constraint as the amount of interpenetration or separation between the parts, it is possible to seek solutions that minimize these values. However, such a solution cannot be determined uniquely. Conversely, a human can narrow down the solution based on human common-sense [10] such as recognizing an object affordance [5]. For example, considering a slider on a plane, the geometric constraint is satisfied when the direction of motion is perpendicular to the normal of the plane. However, common-sense dictates that the direction of motion should align with the functionality of the object itself. Similarly, considering a knob on a plane, the geometric constraint is met when the rotation axis aligns with the normal direction of the plane and can be at any rotation center. However, common-sense suggests that only the center of the knob should be the rotation center. Note that the rotation center of an object, such as a lever, is not the center of the object, making it difficult to determine the center heuristically. Therefore, in this study, we

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propose a system to solve the joint-assignment where a human and machine collaborate to complement each other's strengths and weaknesses (see Figure 1).

We design the optimization based on human common-sense using preferential Bayesian Optimization (PBO), which is a Human-in-the-Loop Bayesian optimization [3, 6, 21] (see the middle part of Figure 1). PBO has been used for the optimization of a design optimization [13, 14, 24]. This study leverages PBO to consider human common-sense. Additionally, to mitigate the burden on a user in considering the geometric constraint, we propose an automatic evaluation where the machine evaluates the candidates based on the geometric constraint (see the upper part of Figure Figure 1). Solely considering the constraint in the automatic evaluation leads to falling into a local solution that deviates from human common-sense. Thus, we propose an escape strategy from the local solution where a user suggests a candidate solution (see the lower part of Figure 1). This strategy is designed based on restart strategies [1, 9].

As a result of a user experiment, we demonstrated that using this system enables joint assignment that satisfies both the human common-sense and the geometric constraint. The result of a questionnaire provided insights that users find it challenging to adjust joint pose and detect interpenetration. In addition, our system has been shown to provide effective support for these issues.

2 GHOJA: GEOMETRIC- AND HUMAN-BASED OBJECT JOINT-ASSIGNMENT

2.1 Overview

In this study, we assume that a 3D model divided into parts is given. The joint-assignment involves determining the joint type, position, and orientation. Since humans can decide the joint type based on common-sense, we allow a user to choose the joint type. The joint-pose is calculated through optimization. In optimization, determining a good initial solution is generally crucial. Since humans can estimate roughly the pose, the user first determines the initial pose. Subsequently, the optimization is performed.

The pose determined by humans is rough from the perspective of the geometric constraint, which ensures that the parts do not interpenetrate or detach each other when the parts are moved using the joint. For example, if the human attaches the translational joint to the slider roughly, the slider interpenetrates into or detaches from a plane as a result of the motion. Therefore, it is necessary to optimize the pose to satisfy the geometric constraint. However, if optimization is solely based on the geometric constraint, it yields a pose that deviates from human common-sense. Hence, it is necessary to optimize the joint-pose by considering not only the geometric constraint but also an objective function that incorporates the human common-sense. The Preferential Bayesian optimization (PBO) is used for the optimization. This allows optimization that incorporates common-sense based on human evaluations. In PBO, multiple parameter candidates are provided, and a human repeatedly selects the worst candidate among these candidates. See [21] for more specific calculation.

2.2 Automatic evaluation based on geometric constraint

In PBO, a human repeatedly selects the worst candidate manually. It can be challenging for a human to assess whether the geometric constraint is satisfied by visual information. Additionally, it generally requires two or three digits of selections to reach convergence [21], making it very time-consuming for a human to continue manual selection until convergence is achieved. Thus, we introduce the automatic evaluation based on the geometric constraint to converge to a solution that satisfies the constraint and reduces the human effort.

In this study, the geometric constraint is represented as the maintenance of contact between the two parts that are jointed. The contact state of two parts can be categorized into three states: the parts come to a stop at the contact surface, push in the direction that causes interpenetration, or detach from each other [10]. For the purpose of the automatic evaluation, we define quantitative geometric metrics d as follows:

$$\begin{cases} d = 0 & (\text{maintenance}) \\ d = d_p & (\text{interpenetration}) \\ d = d_f & (\text{separation}) \end{cases} \quad (1)$$

Here, d_p represents the minimum required distance for the parts to move to resolve the interpenetration between the parts and d_f is defined as the minimum distance between the surfaces of the parts. The value D used during automatic evaluation is defined as the sum of d when the parts are moved at equally spaced intervals. If we divide the range of motion into N equally spaced intervals, and d^i represents the geometric metrics at the i -th time when the parts are moved, then D is defined as:

$$D = \sum_{i=0}^N d^i \quad (2)$$

Note that d^0 and d^N represent the times at the start and end of the range of motion.

During the automatic evaluation, the candidate with the highest D is considered the worst candidate. Achieving $D = 0$ through optimization can be time-consuming. Thus, the automatic evaluation is terminated once D of all candidates falls below a predefined threshold T . At that point, the evaluation moves to the human evaluation stage.

2.3 User's suggestion of candidate solution

When performing optimization based solely on the geometric constraint, there is a risk of converging to solutions that contradict the human common-sense. For example, considering a knob on a plane, the geometric constraint is met when the rotation axis aligns with the normal direction of the plane and can be at any rotation center. The geometric-based optimization falls into such a local minimum that satisfies only the geometric constraint. Generally, when dealing with a local minimum, restart strategies are used, involving sampling random candidates and re-exploring around them [1, 9].

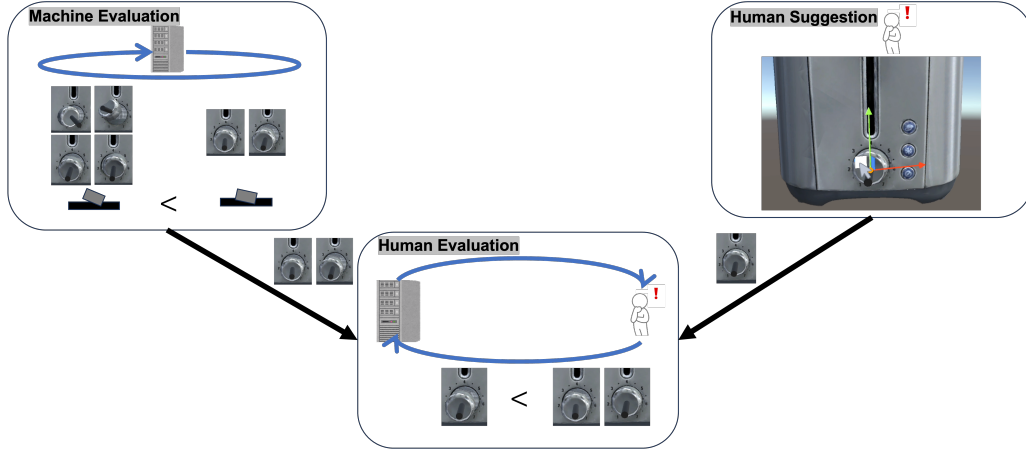


Figure 1: Conceptual diagram of this study. The machine evaluates candidate solutions based on the geometric constraint and presents solutions that meet the constraint to a human (upper part). The human then assesses these solutions based on common-sense (middle part). Additionally, humans have the option to provide their own candidate solutions during the assessment (lower part). Optimization is performed based on these evaluations.

In this study, we design a strategy that allows escaping a local minimum and reaching a solution that meets the human common-sense. While previous studies typically restarted from a random candidate, we propose a strategy restarting from a human suggestion. The suggestion plays a role in accelerating the convergence to solutions that meet the geometric constraint and the human common-sense. This strategy after the suggestion is performed as follows:

- (1) Generate a candidate solution. If the candidate does not satisfy the geometric constraint, proceed to step 2. Otherwise, proceed to step 3.
- (2) Evaluate the suggested candidate as the best one and remove the generated solution. Proceed to step 1.
- (3) Eliminate that suggested candidate and generate a new candidate if the suggested candidate does not satisfy the geometric constraint.

3 IMPLEMENTATION

GHOJA allows users to manipulate the camera to view a 3D model and assign a joint to the 3D model. Figure 2 illustrates the workflow of GHOJA. First, the joint type is determined (Figure 2-(A)). Then, the position and orientation of the joint are manually adjusted (Figure 2-(B)). Subsequently, the user can refine the joint-pose while observing how the parts move (Figure 2-(C)). Finally, joint-pose optimization is performed. The optimization includes three features: automatic evaluation (Figure 2-(D)), human evaluation (Figure 2-(E)), and human suggestion of a candidate solution (Figure 2-(F)). Users are free to choose any of these features. Note that this system is implemented in Unity¹. The camera's operations are designed with reference to Unity's camera movement.

This system provides two types of joints. One is a Prismatic joint, which allows for one degree of freedom in translation only. The other is a revolute joint, which allows for one degree of freedom

in rotation only. Users can select these joints by left-clicking the button in the lower left corner. When a type is selected, the system displays a black prism representing the prismatic joint or a cylinder representing the revolute joint on the screen. The long axis of the prism represents translation, or the axis of rotation (as shown in Figure 2-(A)).

When a joint is left-clicked, an axis for the joint-position adjustment is displayed (Figure 2-(B-1)). Users can translate and rotate the joint (Figure 2-(B-2)). Furthermore, by clicking the "Move" button on the right side of the screen, a slider is displayed in the upper right corner (Figure 2-(C)). This slider allows users to move parts using the current joint-pose. Users can adjust the pose while observing the movement.

The optimization process can be initiated by clicking the "StartPBO" button on the right side of the screen. There are two optimization modes: automatic evaluation and human selection. The button in the center of the screen labeled "Auto" (Figure 2-(D)) activates automatic evaluation, while the button labeled "Manual" (Figure 2-(E)) activates human selection. Users can switch between these modes by pressing the buttons. During the automatic evaluation, if all geometric metrics of the candidate solutions are below the threshold $T = 0.5\text{cm}$, the mode automatically switches to human selection. In human selection, users can view the candidate solutions selected in the dropdown menu in the top left corner and choose the worst candidate from the buttons at the bottom of the screen. The number of candidate solutions is set to 3. Additionally, users can edit the joint-pose and suggest a candidate solution by pressing the "Send new trial" button (Figure 2-(F)). Once a satisfactory candidate is obtained, users can select it from the dropdown menu and end the optimization process by pressing the "Finish" button in the center.

¹<https://unity.com>

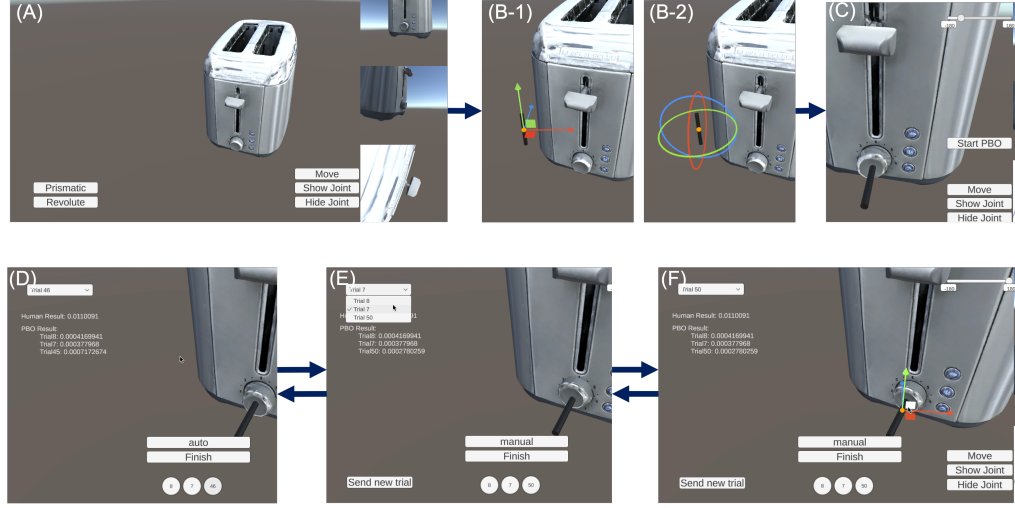


Figure 2: Flowchart of GHJOJA: The upper row represents the features before optimization begins, while the lower row depicts the features after optimization starts. (A): selecting the joint type, (B): manual adjustment of joint-pose, (C): observing part movements in the current joint-pose, (D): automatic evaluation based on the geometric constraint, (E): human evaluation, and (F): human suggestion of candidate solutions. During the optimization, Users are free to choose (D), (E), and (F).

4 USER STUDY

4.1 Hypothesis

In this experiment, we compare three conditions for determining joint-pose: the manual joint-assignment by a human (**Human-baseline**), the automatic assignment based on the geometric constraint (**Machine-baseline**), and the assignment using our proposed method (**GHJOJA**), with a focus on whether they satisfy the geometric constraint and human common-sense. We have two hypotheses: **H1**: Our proposed method is better at satisfying the geometric constraint compared to manual assignment by a human. **H2**: Our proposed method is better at satisfying human common-sense compared to automatic positioning. To test these hypotheses, we design the experiment. Note that for the determination of the joint type, we assume that a human doesn't make a mistake, so we don't verify the accuracy in that aspect.

4.2 Procedure

The experiment was conducted in the following order: a system explanation, a system practice, an actual experiment, and a questionnaire. In the system explanation, participants received instructions on basic operations and the steps involved in the optimization process. In the experiment, participants performed joint-assignment on a 3D model of a toaster obtained by scanning an actual toaster. The toaster's slider and knob were the target parts for joint-assignment. This 3D model is included in Google Scanned Objects [2]. The parts of the toaster were manually separated by the author. Note that in the case of the toaster's slider, the part that should have a groove is filled, creating a completely flat surface. Such irregularities, where grooves are filled in, are common in 3D scans. In this case, the application of the geometric constraint during the optimization doesn't lead to a unique solution. The range of motion of the joint was specified to fixed values.

The manual adjustment and optimization were recorded as Human-baseline and GHJOJA, respectively. Using the result from Human-baseline, the optimization was performed using only the geometric constraint. The results of this optimization were recorded as Machine-baseline. After the joint-assignment, the participants were asked to rank these results in terms of how well they meet human common-sense. They were also encouraged to share free comments about their impressions of the proposed system in the questionnaire.

A total of 12 participants (1 female) took part in the experiment, all from the same organization. The average age of the participants was 29.17 years, with a standard deviation of 7.59. Seven of the participants had experience with 3D computer graphics software, and eight had experience running optimization algorithms.

4.3 Evaluation metrics

To test **H1**, which aims to evaluate whether GHJOJA satisfies the geometric constraint better than Human-baseline, the geometric metrics D is used as the evaluation metrics. To test **H2**, which aims to evaluate whether GHJOJA meets human common-sense better compared to Machine-baseline, we compared these methods by using a human ranking of the joint-pose and the joint-pose error calculated from the ground truth. Friedman's test is conducted, followed by Wilcoxon signed-rank tests. Multiple comparison corrections are applied using the Benjamini-Hochberg method (The data from the experiment did not show normality according to the Anderson-Darling test.)

5 RESULTS

5.1 Results on H1: GHOJA satisfies geometric constraint more than Human-baseline

(A) of the left of Figure 3 shows the results for the toaster slider, and (B) shows the results for the knob. For both the slider and the knob, GHOJA had significantly smaller geometric metrics compared to Human-baseline ($p = 0.006, 0.002 < 0.05$). In addition, for the slider, Machine-baseline was found to have significantly smaller geometric metrics compared to GHOJA ($p = 0.016 < 0.05$). Therefore, **H1** was supported in this experiment. Eight participants mentioned that they had difficulty in discerning the interpenetration and found it helpful that our system automatically resolved the interpenetration. They also found it beneficial that the system automatically adjusted joint-pose, and that better choices were generated.

5.2 Results on H2: GHOJA meets human common-sense more than Machine-baseline

(A) and (B) of the right of Figure 3 show the results of ranking for the slider and knob, respectively. Figure 4-(A) and (B) present the results of errors with respect to the ground truth for the slider and knob. In the case of errors with respect to the ground truth, for the slider, it only includes the orientation errors of the translational axis, while for the knob, it has the orientation error of the rotational axis (B-1) and the position error of the rotational center (B-2). The position of the rotation center was calculated as the intersection between the toaster's plane and the rotation axis. Regarding the ranking, both for the slider and knob, GHOJA outperforms Machine-baseline significantly ($p = 0.042, 0.0004 < 0.05$). For the knob, Human-baseline has a significantly higher ranking than Machine-baseline ($p = 0.0048 < 0.05$). In comparison with the ground truth, GHOJA shows significantly higher accuracy than Machine-baseline for the slider's orientation error and the knob's position error ($p = 0.034, 0.0009 < 0.05$). For the slider and knob's orientation errors, GHOJA has significantly higher accuracy compared to Human-baseline ($p = 0.0004, 0.0024 < 0.05$). Based on the results, we confirmed that **H2** was supported.

Figure 5-(A) compares the results for the slider from participant 9. In Human-baseline, it appears that the parts are moving in the correct direction from the front view. However, the geometric constraint was not satisfied. In Machine-baseline, although the geometric constraint was satisfied, the parts moved in the wrong direction. In contrast, GHOJA achieved results where both the geometric constraint and human common-sense were met. Figure 5-(B) compares the results for the knob from participant 12. In Human-baseline, the geometric constraint was not satisfied. In Machine-baseline, although the geometric constraint was satisfied, the rotation center was incorrectly placed. GHOJA achieved results where both the geometric constraint and human common-sense were met.

6 DISCUSSION

In the experiment, two participants mentioned that the waiting time for the automatic evaluation was long, and the number of trials was high (around 100 trials, including automatic evaluations). Even though it takes a long time, it is important to reduce the error. Indeed, the error of GHOJA is lower than the baselines and related

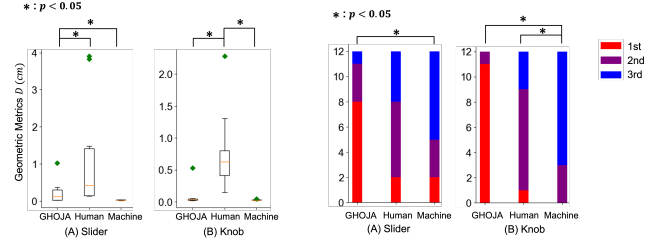


Figure 3: Left: Comparison of the geometric metrics. (A) is for the slider, and (B) is for the knob. The green diamonds represent outliers. Right: Comparison of rankings when arranged in order to meet human common-sense. (A) shows the comparison for the slider, and (B) shows the comparison for the knob.

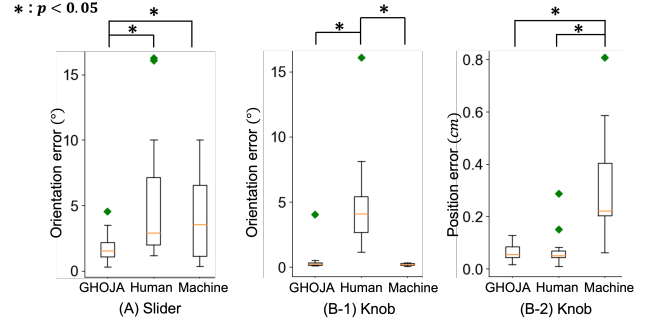


Figure 4: Comparison of errors with respect to the ground truth. (A) shows the comparison of orientation errors for the slider, (B-1) shows the comparison of orientation errors for the knob, and (B-2) shows the comparison of position errors for the knob. The green diamonds represent outliers.

work [11]. However, this problem may decrease the usability and user experience of the system. One possible solution is increasing the number of generated candidates, which leads to accelerating the convergence to the solution. In this case, it is important to allow a user to view all the candidates simultaneously [14] because a user needs to compare many candidates. Additionally, participants enjoyed observing the automatic optimization process. Thus, it is possible to improve the user experience by showing more of the optimization process, such as visualizing the currently explored solution space.

In this study, we proposed a framework to find a solution that satisfies the human common-sense and geometric constraints. This framework can be applied to part assembly [15] as well. Part assembly involves estimating the part pose, assuming that the parts already exist. In this problem, it is necessary to adjust the part pose to satisfy the geometric constraint between the parts. Additionally, the assembly process is influenced by human preference. Therefore, we believe that the proposed framework can be applied to such scenarios. Moreover, in terms of estimation accuracy, our system can be applied in the robot motion planning for manipulating objects with joints, like doors and drawers.

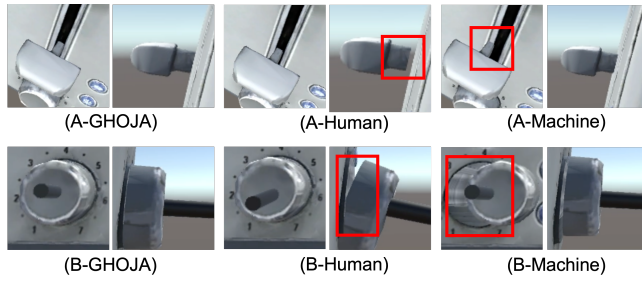


Figure 5: Examples of results obtained from the optimization performed by GHOJA, Human-baseline, and Machine-baseline. (A) shows a comparison of results for participant 9 with the slider, and (B) shows a comparison of results for participant 12 with the knob. In each case, the left image for each method shows a view from above, and the right image shows a side view. The red-enclosed portions highlight areas where the geometric constraint or human common-sense did not meet the requirements.

7 CONCLUSION

In this study, we proposed a system that utilizes Human-in-the-Loop Bayesian optimization, considering the human common-sense and the geometric constraint, to assist users in joint-assignment to 3D models. To reduce the burden of human evaluations, we newly propose a feature where the machine evaluates the candidate solutions automatically based on the geometric constraint. In addition, a human can suggest the candidate solution to escape from a local minimum in the geometric-based optimization. We conducted the experiment for the joint-assignment to the toaster's slider and knob using our system. The results showed that our system satisfies the geometric constraint better than manual joint-assignment by humans. It also demonstrated that our system meets the human common-sense better than the automatic optimization by the machine. Furthermore, user feedback indicated that users found the system effective for identifying the interpenetration and adjusting the joint-pose.

The limitation of this study is that many of the participants had experience with optimization. It is possible that the knowledge about optimization has influenced the results. Future work could investigate whether the results change when the number of participants who have no prior optimization experience increases.

REFERENCES

- [1] Anne Auger and Nikolaus Hansen. 2005. A restart CMA evolution strategy with increasing population size. In *2005 IEEE congress on evolutionary computation*, Vol. 2. IEEE, 1769–1776.
- [2] Laura Downs, Anthony Francis, Nate Koenig, Brandon Kinman, Ryan Hickman, Krista Reymann, Thomas B McHugh, and Vincent Vanhoucke. 2022. Google scanned objects: A high-quality dataset of 3d scanned household items. In *2022 International Conference on Robotics and Automation (ICRA)*. IEEE, 2553–2560.
- [3] Brochu Eric, Nando Freitas, and Abhijeet Ghosh. 2007. Active preference learning with discrete choice data. *Advances in neural information processing systems* 20 (2007).
- [4] Kyle Gao, Yina Gao, Hongjie He, Denning Lu, Linlin Xu, and Jonathan Li. 2022. Nerf: Neural radiance field in 3d vision, a comprehensive review. *arXiv preprint arXiv:2210.00379* (2022).
- [5] James J Gibson. 1977. The theory of affordances. *Hilldale, USA* 1, 2 (1977), 67–82.
- [6] Javier González, Zhenwen Dai, Andreas Damianou, and Neil D Lawrence. 2017. Preferential bayesian optimization. In *International Conference on Machine Learning*. PMLR, 1282–1291.
- [7] Michael Hagenow, Emmanuel Senft, Evan Laske, Kimberly Hambuchen, Terrence Fong, Robert Radwin, Michael Gleicher, Bilge Mutlu, and Michael Zinn. 2022. Registering Articulated Objects With Human-in-the-loop Corrections. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2343–2350.
- [8] Ruizhen Hu, Wenchao Li, Oliver Van Kaick, Ariel Shamir, Hao Zhang, and Hui Huang. 2017. Learning to predict part mobility from a single static snapshot. *ACM Transactions on Graphics (TOG)* 36, 6 (2017), 1–13.
- [9] Xiaohua Hu, Ronald Shonkwiler, and Marcus C Spruill. 1994. Random restarts in global optimization. *Georgia Institute of technology, Atlanta* (1994).
- [10] Katsushi Ikeuchi, Naoki Wake, Kazuhiro Sasabuchi, and Jun Takamatsu. 2023. Semantic constraints to represent common sense required in household actions for multimodal learning-from-observation robot. *The International Journal of Robotics Research* 0, 0 (2023), 02783649231212929. <https://doi.org/10.1177/02783649231212929> arXiv:https://doi.org/10.1177/02783649231212929
- [11] Ajinkya Jain, Stephen Giguere, Rudolf Lioutikov, and Scott Niekum. 2022. Distributional depth-based estimation of object articulation models. In *Conference on Robot Learning*. PMLR, 1611–1621.
- [12] Ajinkya Jain, Rudolf Lioutikov, Caleb Chuck, and Scott Niekum. 2021. Screwnet: Category-independent articulation model estimation from depth images using screw theory. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 13670–13677.
- [13] Yuki Koyama and Masataka Goto. 2022. BO as Assistant: Using Bayesian Optimization for Asynchronously Generating Design Suggestions. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–14.
- [14] Yuki Koyama, Issei Sato, and Masataka Goto. 2020. Sequential gallery for interactive visual design optimization. *ACM Transactions on Graphics (TOG)* 39, 4 (2020), 88–1.
- [15] Yichen Li, Kaichun Mo, Lin Shao, Minhyuk Sung, and Leonidas Guibas. 2020. Learning 3d part assembly from a single image. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VI* 16. Springer, 664–682.
- [16] Liu Liu, Wenqiang Xu, Haoyuan Fu, Sucheng Qian, Qiaojun Yu, Yang Han, and Cewu Lu. 2022. Akb-48: A real-world articulated object knowledge base. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14809–14818.
- [17] Liu Liu, Han Xue, Wenqiang Xu, Haoyuan Fu, and Cewu Lu. 2022. Toward real-world category-level articulation pose estimation. *IEEE Transactions on Image Processing* 31 (2022), 1072–1083.
- [18] Sina Masnadi, JJ LaViola, Jana Pavlasek, Xiaofan Zhu, Karthik Desingh, and Odest Chadwicke Jenkins. 2019. Sketching affordances for human-in-the-loop robotic manipulation tasks. *2nd Robot Teammates Operating in Dynamic, Unstructured Environments (RT-DUNE)* (2019).
- [19] Sina Masnadi, A Vargas, Brian Williamson, and Joseph LaViola. 2020. Affordit!: A tool for authoring object component behavior in virtual reality. *Proceedings of Graphics Interface 2020* (2020).
- [20] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.
- [21] Shion Takeno, Masahiro Nomura, and Masayuki Karasuyama. 2023. Towards Practical Preferential Bayesian Optimization with Skew Gaussian Processes. *arXiv preprint arXiv:2302.01513* (2023).
- [22] Ruicheng Wang, Jialiang Zhang, Jiayi Chen, Yinzheng Xu, Puhao Li, Tengyu Liu, and He Wang. 2023. Dexgraspnet: A large-scale robotic dexterous grasp dataset for general objects based on simulation. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 11359–11366.
- [23] Zeyu Wang, Cuong Nguyen, Paul Asente, and Julie Dorsey. 2023. PointShopAR: Supporting Environmental Design Prototyping Using Point Cloud in Augmented Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.
- [24] Kenta Yamamoto, Yuki Koyama, and Yoichi Ochiai. 2022. Photographic Lighting Design with Photographer-in-the-Loop Bayesian Optimization. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–11.